

The Most Promising BiCRS Pathways in the United Kingdom

Biomass Carbon Removal and Storage (BiCRS) converts waste and residual biomass into durable CO₂ removal — either by capturing and geologically storing the biogenic CO₂ released when biomass is used for energy or fuel, or by locking the carbon away directly. This brief summarises where the UK's most promising opportunities lie, drawn from the interactive **UK BiCRS Atlas** (<https://hausfath.github.io/bicrs-map-uk/>), which resolves feedstock, storage, and best-use recommendations to each of the UK's 41 NUTS-2 / ITL-2 regions.

The Atlas ranks pathways using Frontier's procurement KPIs — **carbon-removal efficiency first, then emissions-avoiding co-products, then wider co-benefits** — screened against feedstock moisture and density, distance and cost to reach CO₂ storage, soil-nutrient status, and whether a facility suitable for retrofit already exists nearby. Purpose-grown energy crops are excluded by design; the focus is genuinely residual and waste biomass.

A note on scope. This brief counts removal from the UK's **own domestic feedstocks** — biomass grown or collected in the UK (crop and forestry residues, manure, municipal and sewage waste). It does **not** count the much larger biogenic CO₂ from power stations that burn **imported** biomass (chiefly wood pellets — Drax, Lynemouth, MGT Teesside). Retrofitting those with capture is a large, separate opportunity, treated on its own in §2, because it depends on continued biomass imports rather than UK supply.

Headline (domestic feedstocks): how much durable removal the UK's own residual biomass supports depends heavily on **what kind of CO₂ storage is available**, so we frame two scenarios:

- **Scenario A — gaseous-CO₂ storage only (~8.1 Mt CO₂/yr).** The UK's developed and in-construction stores (the offshore CCS hubs) accept **gaseous / dense-phase CO₂ only**. This scenario counts just the pathways that capture CO₂ (BECCS, waste-to-energy + CCS, anaerobic digestion + CCS) plus storage-independent biochar. It is the near-term, shovel-ready floor.
- **Scenario B — with salt-cavern injection storage (~14.9 Mt CO₂/yr).** If the UK develops **salt caverns** to store biomass slurry and bio-oil (a recognised method, but no UK site is yet permitted for it), the wet livestock regions and dry-residue regions far from a capture anchor gain a pathway, lifting potential by **~6.8 Mt CO₂/yr (+84%)**.

Both are screening estimates of what a mature, right-sized UK BiCRS sector could deliver from domestic wastes and residues alone, before any dedicated energy crops. The UK is unusually well positioned on the capture side: it pairs a modest but real residue base with the largest offshore CO₂ storage resource in Europe and two storage hubs already in construction.

1. Feedstock availability

The UK is not a biomass-rich country by land area, but its **waste streams are substantial and highly concentrated near population and industry** — favourable for BiCRS, which rewards feedstock that is already aggregated. Recoverable UK totals (Mt/yr, oven-dry except MSW):

Feedstock	Recoverable supply	Character & best fit
Municipal solid waste (total)	~25.7 Mt/yr (~55% biogenic)	Concentrated in cities; already collected → WtE + CCS
Animal manure	~12.0 Mt/yr	Wet, distributed across the livestock west/north → AD + CCS or HTL bio-oil
Agricultural residues	~6.3 Mt/yr	Straw/stover, concentrated in the arable east → BECCS / injection
Human / WWTP biosolids	~1.0 Mt/yr	Wet, at large sewage works → co-digestion / injection
Forestry residues	~0.5 Mt/yr	Small; the UK is import-dependent for wood

Two features stand out. First, **MSW and manure dominate the tonnage** — both wet or heterogeneous streams that suit anaerobic digestion, waste-to-energy, or hydrothermal routes rather than dry combustion. Second, the geography is clean: **arable dry residues cluster in East Anglia and the east**, while **wet manure dominates the livestock west and north** (Wales, South West England, Scotland, Northern Ireland). That split largely determines the recommended pathway region-by-region.

Manure and crop-residue supply are distributed using the **UK June Agricultural Census** (housed cattle/pig/poultry numbers for manure; arable-land area for straw), so recoverable manure tracks where livestock actually are — near-zero in London and the conurbations, concentrated in Northern Ireland, West Wales, the South West, North Yorkshire and southern Scotland.

2. Retrofit opportunities

The UK's strongest near-term advantage is a **large existing biomass and waste-combustion fleet that can be retrofitted with capture**, avoiding greenfield build. The Atlas maps 26 candidate biogenic point sources:

- **Biomass power (11 sites, ~18.9 Mt biogenic CO₂/yr of throughput) — mostly imported biomass.** The large stations run predominantly on **imported wood pellets: Drax** (~12 Mt/yr biogenic CO₂ — the single largest BECCS opportunity in Europe), **Lynemouth** (~2.7), and **MGT Teesside** (~2.0) together account for ~16.7 Mt/yr. Retrofitting them with capture is a **large, concentrated, retrofit-ready removal opportunity — but it sits outside the domestic-feedstock headline** because it depends on continued sustainable-biomass imports (a distinct supply, cost and sustainability question). A cluster of smaller ~40 MW plants (Snetterton, Sleaford, Brigg, Kent/Sandwich, Ince/Mersey) burns largely **domestic** straw and waste wood and is reflected in the BECCS line of the scenario tables (East Anglia / Essex).
- **Waste-to-energy (13 sites, ~3.0 Mt biogenic CO₂/yr).** The urban EfW fleet — Runcorn, Cory Riverside/Belvedere, Edmonton, Lakeside, Earls Gate (Scotland) — anchors **WtE + CCS** across the English cities and central belt.
- **Anaerobic digestion (existing UK biogas cluster).** Hundreds of farm and food-waste digesters already operate; retrofitting biogas-upgrading CO₂ vents is among the cheapest capture available and anchors **AD + CCS** across the livestock regions.

Crucially, these anchors sit close to the **CO₂ storage hubs now in construction**: the **Northern Endurance Partnership / East Coast Cluster** (Teesside–Humber) and **Liverpool Bay / HyNet** (Mersey), with **Acorn** (Scotland) and **Viking** (Humber) planned. In the Atlas, **all 41 UK regions have "good" (economically reachable) access to gaseous-CO₂ storage** — a far higher share than most of Europe — because these offshore stores are within short transport reach. Modelled delivered CO₂-transport cost is low (~35–65/tCO₂, median 50), reflecting short routes to the North Sea. This is what makes the CO₂-capture pathways (Scenario A) robust across the whole country.

3. Total CO₂ removal potential by pathway — two storage scenarios

The decisive question for UK BiCRS is **what the CO₂ can be stored as**. Capturing CO₂ as a gas and piping it to an offshore reservoir (BECCS, WtE+CCS, AD+CCS) uses storage the UK is already building. Sequestering **biomass slurry or bio-oil** instead requires a store that takes liquids/solids — in the UK, that means **salt caverns**, which are not yet developed or permitted for this purpose. The two scenarios below bracket that choice.

Scenario A — gaseous-CO₂ storage only (near-term floor: ~8.1 Mt CO₂/yr)

Only CO₂-capture pathways (routing to the offshore hubs) plus storage-independent biochar.

Pathway	Regions	CDR potential	CDR efficiency	Indicative cost
WtE + CCS (capture on energy-from-waste)	18	4.1 Mt CO ₂ /yr	~55%	~\$100–200/t

Pathway	Regions	CDR potential	CDR efficiency	Indicative cost
AD + CCS (biogas upgrading capture)	9	2.3 Mt CO ₂ /yr	~37%	\$145–300/t
BECCS (biomass power + capture)	2	1.6 Mt CO ₂ /yr	~80%	\$200–225/t
Biochar (storage-independent)	1	0.1 Mt CO ₂ /yr	~high	~\$100–200/t
No viable pathway	11	—	—	—
Total	41	~8.1 Mt CO ₂ /yr		

The 11 "no viable pathway" regions are the wet livestock areas without an existing digester — Northern Ireland, Devon, Dorset & Somerset, Cornwall, Cumbria, Teesside, and most of Scotland. Wet manure cannot be combusted or buried, and with no gaseous-CO₂-only route for it, it has *no* removal option here unless anaerobic digestion is built (moving it into AD+CCS) or injection storage becomes available (Scenario B).

Scenario B — with salt-cavern injection storage (~14.9 Mt CO₂/yr)

Developing salt caverns for biomass-slurry / bio-oil injection adds two pathways and rescues the livestock regions. Injection becomes the largest single pathway.

Pathway	Regions	CDR potential	CDR efficiency	Indicative cost
Biomass waste injection (slurry → salt cavern)	8	5.6 Mt CO ₂ /yr	~90%	\$125–285/t
WtE + CCS	18	4.1 Mt CO ₂ /yr	~55%	~\$100–200/t
AD + CCS	9	2.3 Mt CO ₂ /yr	~37%	\$145–300/t
BECCS	2	1.6 Mt CO ₂ /yr	~80%	\$200–225/t
HTL bio-oil (bio-crude → salt cavern)	4	1.3 Mt CO ₂ /yr	~50%	\$200–400/t
Total	41	~14.9 Mt CO ₂ /yr		

Salt caverns unlock ~6.8 Mt CO₂/yr (+84%), almost entirely in the wet livestock regions. But this storage is **prospective** — the UK has substantial salt-cavern resource (Cheshire, Teesside, East Yorkshire, Lancashire, Dorset, and Islandmagee in Northern Ireland) and bio-oil/biomass storage in salt caverns is a credited method, yet **no UK site is developed or permitted for it today**. Treat Scenario B as developable potential contingent on that storage being established.

Domestic feedstocks vs. total including imported-biomass retrofits

The two scenarios above are **domestic-feedstock** potential. Separately, capturing the biogenic CO₂ from the UK's **imported-biomass power stations** (Drax, Lynemouth, MGT Teesside) would add on the order of ~17 Mt CO₂/yr of BECCS — comparable to, or larger than, the entire domestic potential. Combining the two views:

View	Removal potential	Notes
Domestic feedstocks, gaseous-CO ₂ storage (Scenario A)	~8.1 Mt CO ₂ /yr	Near-term, UK-grown/collected biomass, existing storage
Domestic feedstocks, with salt-cavern injection (Scenario B)	~14.9 Mt CO ₂ /yr	Adds prospective injection storage
+ Imported-biomass power-station retrofits	+~17 Mt CO ₂ /yr	Drax/Lynemouth/MGT; depends on continued biomass imports
Total including imported-biomass retrofits	~25–32 Mt CO ₂ /yr	Domestic (A or B) + imported-biomass BECCS

The imported-biomass line is the UK's single largest near-term BECCS lever (Drax alone rivals the whole domestic total) and its stations already sit on the storage clusters — but because it relies on imported wood pellets, its scale is a **policy and supply-chain choice**, not a property of UK land. Frontier's interest is durable, additional, high-quality removal; the domestic-feedstock figures are the more directly UK-controllable base, with imported-biomass retrofit as a large optional addition.

How to read this. Two points for prioritisation:

- **Biggest and lowest-regret, in both scenarios: WtE + CCS.** ~4.1 Mt/yr across 18 regions — MSW is the UK's largest biogenic waste stream, concentrated in cities that already run energy-from-waste plants. It removes carbon *and* decarbonises waste the UK must manage anyway, and retrofits an existing urban fleet on storage that already exists. **BECCS** is the highest-efficiency capture route (~80%); **Drax alone** represents removal of the same order as either scenario's total if fully captured — the single most consequential UK decision.
- **The livestock question is a storage question.** Whether the UK's abundant wet manure (the west and north) can do BiCRS at scale hinges on either building anaerobic digestion (AD+CCS, gaseous route) or developing salt-cavern injection. Absent both, those ~11 regions have no near-term pathway.

Bottom line for the UK

The UK's BiCRS advantage is **an existing combustion/AD fleet to retrofit plus the best offshore CO₂ storage access in Europe**. From **domestic feedstocks**, ~8 Mt/yr of CO₂-capture removal (WtE+CCS, BECCS, AD+CCS) is robust and near-term; a further ~7 Mt/yr sits in the wet livestock regions and **hinges on a storage decision** — developing salt caverns for biomass/bio-oil injection (or building out anaerobic digestion) converts that latent potential into deliverable removal. Layered on top, retrofitting the **imported-biomass power stations** (Drax et al.) offers ~17 Mt/yr more, the single largest near-term lever but one contingent on continued sustainable-biomass imports. The fastest, highest-quality domestic tonnes come from capturing CO₂ on the plants already clustered around the Teesside/Humber, Mersey, and central-belt hubs; the strategic questions are whether to unlock injection storage (domestic upside) and how far to lean on imported-biomass BECCS. Explore any region's rationale, ranked alternatives, and delivered costs in the interactive Atlas.

Source: UK BiCRS Atlas (<https://hausfath.github.io/bicrs-map-uk/>), the UK subset of Frontier's BiCRS Atlas. Manure and crop residues distributed by the UK June Agricultural Census (Eurostat Farm Structure Survey); MSW and biosolids by population; forestry from JRC ENSPRESO — all scaled to UK country totals. Gaseous-CO₂ storage from JRC CO₂StoP and curated UK CCS projects; salt-cavern injection sites (Scenario B) are curated and treated as prospective (developable, not yet permitted for bio-oil/biomass storage). Facilities curated with capacity-estimated biogenic CO₂. All figures are screening-level estimates carrying material uncertainty and are not a substitute for project-level diligence. Grounded in Frontier's BiCRS purchasing point of view.